

CVAI = Cranial Vault Asymmetry Index. It quantifies *asymmetry*, so it belongs to plagiocephaly. Scaphocephaly is captured by the cranial index, not by CVAI.

The three deformities are three distinct geometric properties, and each needs its own metric:

Deformity	Geometric abnormality	Metric	Direction
Plagiocephaly	Asymmetry — parallelogram distortion	CVAI	↑
Brachycephaly	Too wide for its length	CI	↑
Scaphocephaly / dolichocephaly	Too narrow for its length	CI	↓

CI is computed from two orthogonal diameters through the midline, so it is **blind to asymmetry** — a markedly plagiocephalic head can have a perfectly normal CI. CVAI is computed from two oblique diagonals, so it is **blind to overall proportion**. They are independent, which is why a calculator needs both: combined deformational brachycephaly *and* plagiocephaly is the commonest clinical presentation.

How CVAI is measured. Two diagonals taken at 30° either side of the midline (some protocols use 40°), then:

$$\text{CVAI (\%)} = (\text{longer diagonal} - \text{shorter diagonal}) \div \text{shorter diagonal} \times 100$$

Conventional bands: <3.5% normal, 3.5–6.25% mild, 6.25–8.75% moderate, >8.75% severe. The related **CVA** is simply the absolute difference in millimetres, unnormalised.

One trap if you're implementing it: **the denominator convention is not uniform**. Some publications divide by the longer diagonal rather than the shorter, which yields systematically smaller values and shifts every threshold. Also fix your diagonal angle — 30° and 40° protocols are not interchangeable. Both should be stated on the output alongside the reference set, for the same reason you're naming the CI reference.

Two clinical points worth encoding:

A normal CVAI does not reassure in scaphocephaly. Sagittal synostosis is a symmetric deformity, so CVAI stays normal while CI falls. If a user relies on CVAI as a general "head shape abnormal?" screen, this is the case they will miss.

CVAI cannot distinguish deformational from synostotic plagiocephaly. Both raise it. The discrimination is morphological, not metric: deformational posterior plagiocephaly gives a

parallelogram — ipsilateral occipital flattening with *ipsilateral* frontal bossing and anterior ear displacement. Lambdoid synostosis gives a **trapezoid** — with *contralateral* frontal bossing and an ipsilateral mastoid bulge. No index captures that; it needs examination.

So your calculator wants four measurements — OFD, BPD, and the two obliques — feeding two independent indices that answer different questions.

Happy to extend the vertex-view diagram to show the 30° diagonals and the parallelogram-versus-trapezoid distinction, if that would be useful for the tool's help screen.